View and update cluster web configurations

ONTAP 9.11.1 REST API reference

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View and update cluster web configurations

Cluster web endpoint overview

Overview

You can use this API to update web services configurations and to retrieve current configurations.

Retrieving the current web services configuration

The cluster web GET API retrieves the current cluster-wide configuration.

Updating the current web services configuration

The cluster web PATCH API updates the current cluster-wide configuration.

Once updated, ONTAP restarts the web services to apply the changes.

When updating the certificate, the certificate UUID of an existing certificate known to ONTAP must be provided. The certificate must be of type "server".

A "client-ca" certificate must be installed on ONTAP to enable "client_enabled".

The following fields can be used to update the cluster-wide configuration:

- enabled
- http_port
- https_port
- http_enabled
- csrf.protection_enabled
- csrf.token.concurrent_limit
- csrf.token.idle_timeout
- csrf.token.max_timeout
- certificate.uuid
- client_enabled
- ocsp_enabled

Examples

Retrieving the cluster-wide web services configuration

```
# API:
GET /api/cluster/web

# The call:
curl -X GET "https://<mgmt-ip>/api/cluster/web" -H "accept:
application/hal+json"

# The response:
{
"enabled": true,
"http_port": 80,
"https_port": 443,
"state": "online",
"http_enabled": false,
"csrf": {
  "protection_enabled": true,
  "token": {
    "concurrent_limit": 500,
    "idle_timeout": 900,
    "max_timeout": 0
  }
},
"certificate": {
  "uuid": "a3bb219d-4382-1fe0-9c06-1070568ea23d",
  "name": "cert1",
  "_links": {
    "self": {
      "href": "/api/security/certificates/a3bb219d-4382-1fe0-9c06-
1070568ea23d"
    }
  }
},
"client_enabled": false,
"ocsp_enabled": false,
"_links": {
  "self": {
    "href": "/api/cluster/web"
  }
}
}
```

Updating the cluster-wide web services configuration

```
# The API:
PATCH /api/cluster/web

# The call:
curl -X PATCH "https://<mgmt-ip>/api/cluster/web" -d '{ "https_port": 446,
"csrf": { "token": { "concurrent_limit": 600 } } }' -H "accept:
application/hal+json"

# The response:
HTTP/1.1 202 Accepted
Date: Fri, 28 May 2021 09:36:43 GMT
Server: libzapid-httpd
Cache-Control: no-cache,no-store,must-revalidate
Content-Length: 189
Content-Type: application/hal+json
```

Retrieve the web services configuration

GET `/cluster/web`

Introduced In: 9.10

Retrieves the web services configuration.

Parameters

Name	Type	In	Required	Description
fields	array[string]	query	False	Specify the fields to return.

Response

```
Status: 200, Ok
```

Name	Type	Description
_links	_links	
certificate	certificate	Certificate used by cluster and node management interfaces for TLS connection requests.
client_enabled	boolean	Indicates whether client authentication is enabled.

Name	Type	Description
csrf	csrf	
enabled	boolean	Indicates whether remote clients can connect to the web services.
http_enabled	boolean	Indicates whether HTTP is enabled.
http_port	integer	HTTP port for cluster-level web services.
https_port	integer	HTTPS port for cluster-level web services.
ocsp_enabled	boolean	Indicates whether online certificate status protocol verification is enabled.
per_address_limit	integer	The number of connections that can be processed concurrently from the same remote address.
state	string	State of the cluster-level web services.
wait_queue_capacity	integer	The maximum size of the wait queue for connections exceeding the per-address-limit.

Example response

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "certificate": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "cert1",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "csrf": {
    "token": {
      "concurrent_limit": 120
    }
  },
  "per_address_limit": 42,
  "state": "offline"
}
```

Error

```
Status: Default, Error
```

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

certificate

Certificate used by cluster and node management interfaces for TLS connection requests.

Name	Type	Description
_links	_links	
name	string	Certificate name
uuid	string	Certificate UUID

token

Name	Type	Description
concurrent_limit	integer	Maximum number of concurrent CSRF tokens.
idle_timeout	integer	Time for which an unused CSRF token is retained, in seconds.
max_timeout	integer	Time for which an unused CSRF token, regardless of usage is retained, in seconds.

csrf

Name	Type	Description
protection_enabled	boolean	Indicates whether CSRF protection is enabled.
token	token	

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

Update the web services configuration

PATCH `/cluster/web`

Introduced In: 9.10

Updates the web services configuration.

Related ONTAP commands

- `system services web modify`

Parameters

Name	Type	In	Required	Description
return_timeout	integer	query	False	The number of seconds to allow the call to execute before returning. When doing a POST, PATCH, or DELETE operation on a single record, the default is 0 seconds. This means that if an asynchronous operation is started, the server immediately returns HTTP code 202 (Accepted) along with a link to the job. If a non-zero value is specified for POST, PATCH, or DELETE operations, ONTAP waits that length of time to see if the job completes so it can return something other than 202. • Default value: 1 • Max value: 120 • Min value: 0

Request Body

Name	Type	Description
_links	_links	
certificate	certificate	Certificate used by cluster and node management interfaces for TLS connection requests.
client_enabled	boolean	Indicates whether client authentication is enabled.
csrf	csrf	

Name	Type	Description
enabled	boolean	Indicates whether remote clients can connect to the web services.
http_enabled	boolean	Indicates whether HTTP is enabled.
http_port	integer	HTTP port for cluster-level web services.
https_port	integer	HTTPS port for cluster-level web services.
ocsp_enabled	boolean	Indicates whether online certificate status protocol verification is enabled.
per_address_limit	integer	The number of connections that can be processed concurrently from the same remote address.
state	string	State of the cluster-level web services.
wait_queue_capacity	integer	The maximum size of the wait queue for connections exceeding the per-address-limit.

Example request

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "certificate": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "cert1",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "csrf": {
    "token": {
      "concurrent_limit": 120
    }
  },
  "per_address_limit": 42,
  "state": "offline"
}
```

Response

```
Status: 200, Ok
```

Response

```
Status: 202, Accepted
```

Error

```
Status: Default
```

ONTAP Error Response Codes

Error Code	Description
9830406	Reconfiguration of the web services failed.
9830407	The web services failed to restart.
9830408	Reconfiguration and/or restart of the web services failed.
9830442	Client authentication cannot be enabled without a client ca certificate.
9830463	The cluster must be fully upgraded before modifying this resource.
9830464	HTTP cannot be enabled when FIPS is also enabled.
9830483	The CSRF token timeout is invalid.
9830484	The maximum concurrent CSRF token count cannot be lower than 100.
9830485	The CSRF idle timeout cannot be greater than the CSRF absolute timeout.
9830486	CSRF requires an effective cluster version of 9.7 or later.
9830487	The HTTP and HTTPS ports must not have the same value.
9830488	The certificate is not a "server" certificate.
9830489	The certificate does not exist for the given SVM.

Name	Type	Description
error	error	

Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

Definitions

See Definitions

href

Name	Type	Description
href	string	

_links

Name	Type	Description
self	href	

certificate

Certificate used by cluster and node management interfaces for TLS connection requests.

Name	Type	Description
_links	_links	
name	string	Certificate name
uuid	string	Certificate UUID

token

Name	Type	Description
concurrent_limit	integer	Maximum number of concurrent CSRF tokens.
idle_timeout	integer	Time for which an unused CSRF token is retained, in seconds.
max_timeout	integer	Time for which an unused CSRF token, regardless of usage is retained, in seconds.

csrf

Name	Type	Description
protection_enabled	boolean	Indicates whether CSRF protection is enabled.
token	token	

web

Name	Type	Description
_links	_links	
certificate	certificate	Certificate used by cluster and node management interfaces for TLS connection requests.
client_enabled	boolean	Indicates whether client authentication is enabled.
csrf	csrf	
enabled	boolean	Indicates whether remote clients can connect to the web services.
http_enabled	boolean	Indicates whether HTTP is enabled.
http_port	integer	HTTP port for cluster-level web services.
https_port	integer	HTTPS port for cluster-level web services.
ocsp_enabled	boolean	Indicates whether online certificate status protocol verification is enabled.
per_address_limit	integer	The number of connections that can be processed concurrently from the same remote address.
state	string	State of the cluster-level web services.
wait_queue_capacity	integer	The maximum size of the wait queue for connections exceeding the per-address-limit.

error_arguments

Name	Type	Description
code	string	Argument code
message	string	Message argument

error

Name	Type	Description
arguments	array[error_arguments]	Message arguments
code	string	Error code
message	string	Error message
target	string	The target parameter that caused the error.

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